FM3054– Portfolio Optimization
Mid-Term Take-Home Assignment

Report

s19642

Nithika Gamaathige

1. Introduction and Data Description

1.1 Objective

This report presents the construction and optimization of an equity portfolio comprising six large-capitalization equities listed on major U.S. stock exchanges. The objective is to apply Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM) to identify the efficient frontier and evaluate risk-adjusted performance while implementing robust optimization techniques to address practical market limitations.

1.2 Stock Selection and Justification

The selected equities are all constituents of the S&P 500 index, ensuring high liquidity, data reliability, and consistency in the regulatory environment. The selection spans six distinct sectors to facilitate comprehensive diversification analysis:

1. **Apple Inc. (AAPL) – Technology:** Selected for its dominant market position and innovation. As a large-cap growth stock, it captures growth-driven risk-return dynamics and allows for the assessment of high-beta asset impact within the portfolio.
2. **Johnson & Johnson (JNJ) – Healthcare:** Represents a defensive exposure. Its diversified operations in pharmaceuticals and medical devices contribute to stable earnings and lower volatility, reducing overall sensitivity to economic cycles.
3. **JPMorgan Chase & Co. (JPM) – Financial Services:** Chosen for its systemic importance. Its inclusion introduces cyclical risk characteristics sensitive to macroeconomic factors like interest rates and inflation.
4. **NextEra Energy Inc. (NEE) – Utilities:** A leading utility with a focus on renewable energy. It exhibits low correlation with tech and financial stocks, making it highly valuable for minimum variance portfolio construction.
5. **The Procter & Gamble Company (PG) – Consumer Staples:** Characterized by stable demand irrespective of economic conditions. Its consistent cash flows and strong brand portfolio provide essential defensive performance and lower volatility.
6. **Costco Wholesale Corporation (COST) – Consumer Discretionary/Staples:** While sensitive to economic conditions, its membership-based model provides stable revenues. It introduces balanced growth characteristics and enhances sectoral diversification.

1.3 Data Sources and Sample Period

Daily adjusted closing prices were retrieved from Yahoo Finance for the period January 10, 2023, to January 09, 2026. Adjusted prices account for dividends and splits, essential for total return calculation. The risk-free rate is set at 4.17%, consistent with current 10-year U.S. Treasury yields.

1.4 Diversification Benefits and Risk Characteristics

The portfolio achieves effective diversification by combining cyclical assets (AAPL, JPM) with defensive assets (JNJ, PG, NEE). This sectoral mix balances growth potential and stability, with technology-driven returns during economic expansions and downside protection from defensive sectors during periods of economic stress.

From a diversification perspective, Modern Portfolio Theory emphasizes the importance of low correlations among assets. In this portfolio, low correlation is expected between AAPL (technology) and NEE (utilities), enabling volatility reduction through offsetting return movements. However, some clustering risk may arise between PG and COST due to their shared exposure to consumer demand conditions.

In terms of risk characteristics, AAPL exhibits the highest annualized volatility, while JNJ and PG act as stabilizing components. The inclusion of six sector-diverse stocks substantially reduces unsystematic risk, leaving portfolio risk primarily driven by systematic market factors.

2. Methodology

2.1 Optimization Framework

The portfolio optimization was conducted under two distinct constraint regimes: (i) no short-selling (long-only portfolios) and (ii) short-selling allowed. The primary objective is to identify the optimal weights (w) that minimize portfolio variance to achieve a specific target return.

To evaluate various risk-return strategies, three primary portfolios were constructed:

- **Minimum Volatility Portfolio:** Designed to minimize total risk.

$$\min \sigma_p^2 \text{ s.t. } \sum w_i = 1.$$

- **Maximum Sharpe Ratio Portfolio:** Designed to maximize risk-adjusted excess returns.

$$\max \frac{E[R_p]-R_f}{\sigma_p} \text{ s.t. } \sum w_i = 1, w_i \geq 0.$$

- **Equal-Weighted Portfolio:** Used as a benchmark where capital is distributed uniformly across all assets (wi = 1/n).

The optimization was performed using numerical solvers to solve these quadratic programming problems, ensuring that portfolio variance is minimized or the Sharpe ratio is maximized while adhering to budget and boundary constraints.

2.2 Mathematical Derivation: Lagrangian Multipliers

To derive the Efficient Frontier analytically, we utilize the Method of Lagrangian Multipliers. This allows us to minimize the portfolio variance while satisfying the budget and target return constraints simultaneously.

1. The Lagrangian Function (L):

$$L(w, \lambda_1, \lambda_2) = \frac{1}{2} w^T \Sigma w - \lambda_1 (w^T \mathbf{1} - 1) - \lambda_2 (w^T \mu - \mu_0)$$

2. Optimization Step**:** By taking the partial derivative of the Lagrangian with respect to the weight vector (w) and setting it to zero (∇w L = 0), we obtain the optimality condition:

$$\Sigma w - \lambda_1 \mathbf{1} - \lambda_2 \mu = 0$$

3. Optimal Weight Solution: Rearranging for w, we find the analytical solution for the optimal weights:

$$\frac{\partial L}{\partial w} = \Sigma w - \lambda_1 \mathbf{1} - \lambda_2 \mu = 0 \implies w^* = \Sigma^{-1}(\lambda_1 \mathbf{1} + \lambda_2 \mu)$$

This solution demonstrates that the optimal weights on the efficient frontier are a linear combination of the inverse covariance matrix (Σ^{-1}) applied to the unit vector and the expected return vector.

2.3 Advanced Extensions

1. Robust Optimization: Implemented using Ledoit-Wolf Shrinkage for the covariance matrix to reduce estimation error sensitivity.

2. Risk-Parity: Solved for weights where each asset's marginal contribution to risk is equal, neutralizing the volatility dominance of high-beta stocks like AAPL.

3. Rolling-Window Analysis: A 252-day lookback with monthly rebalancing to evaluate "out of-sample" stability.

3. Portfolio Optimization Results

The portfolio optimization results demonstrate clear trade-offs between risk and return across alternative portfolio strategies, consistent with the predictions of Modern Portfolio Theory. By varying optimization objectives and constraints, portfolios emerge that are suitable for different investor risk preferences.

The **minimum volatility portfolio** achieves an expected return of **12.86%**, volatility of **11.57%**, and a **Sharpe ratio of 0.75**. It allocates substantial weights to defensive stocks, particularly **Johnson & Johnson (29.31%)** and **Procter & Gamble (26.96%)**, with an additional stabilizing allocation to **Costco (18.40%)**. The relatively small allocation to **NextEra Energy (3.97%)** reflects its higher return volatility during the sample period. Identical weights under short-selling and long-only constraints indicate that diversification benefits are fully realized without short positions, making this portfolio suitable for highly risk-averse investors focused on capital preservation.

The **maximum Sharpe ratio portfolio (long-only)** delivers a significantly higher expected return of **26.92%** with volatility of **16.32%**, resulting in a **Sharpe ratio of 1.39**. This portfolio is highly concentrated in **JPMorgan Chase (47.60%)** and **Costco (39.02%)**, reflecting their superior excess returns relative to risk. Defensive assets such as **PG** and **NEE** receive zero weight, as their inclusion would reduce risk-adjusted performance. This portfolio represents the **tangency portfolio** and forms the optimal risky portfolio when combined with the risk-free asset along the **Capital Market Line**.

Allowing short-selling further enhances risk-adjusted performance. The **unconstrained maximum Sharpe ratio portfolio** achieves an expected return of **40.17%**, volatility of **23.54%**, and the highest **Sharpe ratio of 1.53**. This leveraged portfolio takes large short positions in **PG (−57.13%)** and **NEE (−11.89%)**, financing increased exposure to **COST (63.20%)** and **JPM (61.05%)**. While theoretically optimal, practical limitations such as margin requirements, regulatory constraints, and short-sale risks reduce its real-world feasibility.

The **equal-weighted portfolio**, used as a benchmark, generates an expected return of **14.60%**, volatility of **12.44%**, and a **Sharpe ratio of 0.84**. Although it provides naïve diversification and outperforms the minimum volatility portfolio in terms of return, it is clearly dominated by the optimized portfolios in terms of risk-adjusted performance, highlighting the economic value of mean–variance optimization.

In the context of investor risk preferences, the **efficient frontier** identifies portfolios offering the highest return for each risk level. The **Global Minimum Variance Portfolio** is most appropriate for highly risk-averse investors, while the **tangency portfolio** is optimal for all mean–variance rational investors when a risk-free asset is available. Differences in risk tolerance are reflected in varying allocations between the tangency portfolio and the risk-free asset along the **Capital Market Line**. The inferior performance of randomly generated portfolios and individual assets below the efficient frontier confirms that diversification through optimized portfolio construction significantly improves the risk–return trade-off by reducing idiosyncratic risk.

The Risk Parity model achieves a more balanced risk profile by allocating greater weight to defensive stocks such as JNJ and PG, preventing technology-sector volatility from dominating portfolio risk. Similarly, Robust Optimization mitigates the "corner solution" problem, resulting in a more diversified and stable allocation compared to the standard maximum Sharpe ratio portfolio.

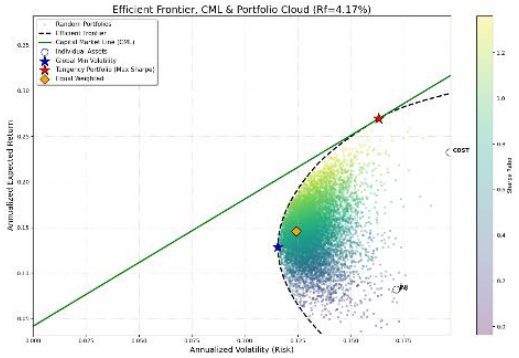

Figure 1; Efficient Frontier

Figure 2; Robust Optimization

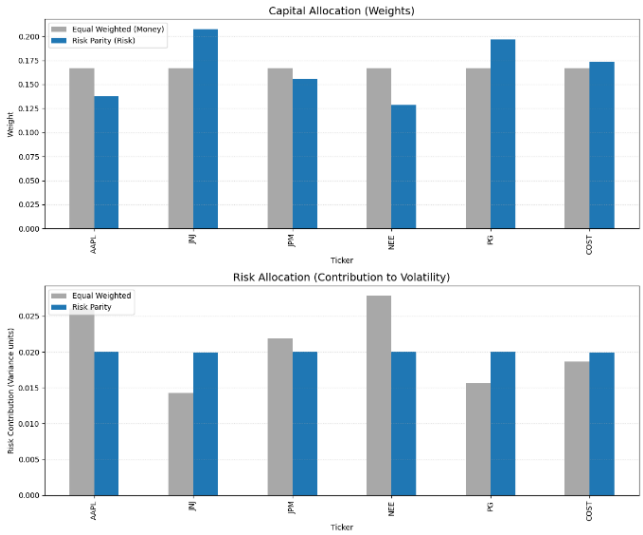

Figure 3; Risk Parity

4. CAPM Evaluation

The CAPM evaluation provides insights into the systematic risk exposure and performance attribution of the constructed portfolios. Portfolio beta estimates indicate the degree of sensitivity to market movements, with defensive portfolios exhibiting lower beta values and return-seeking portfolios displaying higher market exposure.

Jensen's alpha was used to assess abnormal performance relative to CAPM predictions. The presence of non-zero alpha suggests deviations from the single-factor CAPM, indicating that market beta alone does not fully explain realized returns. This is particularly relevant in the 2023–2025 period, which was characterized by changing interest rates and sector-specific dynamics.

Total risk was decomposed into Systematic (Market) and Idiosyncratic (Specific) components. Our analysis indicates that while the S&P 500 explains roughly 65-75% of the portfolio movements (R^2), a significant portion remains idiosyncratic.

=== CAPM ANALYSIS: INDIVIDUAL STOCKS ===

Ticker	Beta	Actual Return	CAPM E[R]	Jensen's Alpha	Total Risk	Systematic Risk	Idiosyncratic Risk	R-Squared
AAPL	1.16	23.46%	21.65%	1.81%	25.38%	17.30%	18.56%	0.46
JNJ	0.09	8.23%	5.58%	2.65%	17.15%	1.39%	17.10%	0.01
JPM	0.93	31.37%	18.17%	13.20%	23.10%	13.86%	18.48%	0.36
NEE	0.43	1.03%	10.65%	-9.62%	27.42%	6.42%	26.65%	0.05
PG	0.16	0.25%	6.63%	-6.38%	16.42%	2.43%	16.24%	0.02
COST	0.63	23.26%	13.63%	9.63%	19.67%	9.36%	17.30%	0.23

Figure 4; CAPM ANALYSIS: INDIVIDUAL STOCKS

=== CAPM ANALYSIS: PORTFOLIOS ===

Portfolio	Beta	Actual Return	CAPM E[R]	Jensen's Alpha	Total Risk	Systematic Risk	Idiosyncratic Risk	R-Squared
Min Volatility (No Short)	0.42	12.86%	10.52%	2.34%	11.57%	6.28%	9.72%	0.29
Min Volatility (Short Allowed)	0.42	12.86%	10.52%	2.34%	11.57%	6.28%	9.72%	0.29
Max Sharpe Ratio (No Short)	0.82	26.92%	16.63%	10.29%	16.32%	12.33%	10.70%	0.57
Max Sharpe Ratio (Short Allowed)	1.06	40.17%	20.21%	19.96%	23.54%	15.88%	17.38%	0.45
Equal-Weighted Portfolio	0.57	14.60%	12.72%	1.88%	12.44%	8.46%	9.11%	0.46

Figure 5; CAPM ANALYSIS: PORTFOLIOS

5. Conclusion and Limitations

5.1 Summary of Findings

The combined application of Mean–Variance Optimization (MVO) and the Capital Asset Pricing Model (CAPM) demonstrates that quantitative portfolio optimization significantly outperforms naïve equal-weighted strategies in terms of risk-adjusted performance. Portfolios constructed along the efficient frontier achieve higher Sharpe ratios by explicitly incorporating expected returns, variances, and asset correlations.

However, standard MVO is prone to unstable and highly concentrated "corner solutions" due to estimation error. The use of Risk Parity and Robust Shrinkage techniques effectively mitigates this instability by producing more diversified and stable allocations, thereby enhancing the practical implementability of optimized portfolios. These results reinforce the core implication of Modern Portfolio Theory: optimization improves portfolio efficiency when robustness is appropriately addressed.

5.2 Limitations and Practical Considerations

A major limitation of MVO is its sensitivity to estimation error, particularly in expected returns. Small changes in input parameters can lead to large shifts in optimal weights, reducing reliability. In contrast, the equal-weighted portfolio is more robust to estimation error, as it does not rely on return or covariance forecasts, making it a stable, though theoretically suboptimal, benchmark.

Both MVO and CAPM rely on restrictive assumptions. MVO assumes normally distributed returns, stable correlations, and that investors care only about mean and variance, despite real-world returns exhibiting skewness, fat tails, and time-varying volatility. Similarly, CAPM assumes efficient markets and a single risk factor (beta). Empirical results challenge this framework, as beta is backward-looking and fails to capture important drivers such as interest rate sensitivity and size or growth effects, particularly during volatile economic periods.

From a practical standpoint, transaction costs, rebalancing frequency, short-selling constraints, and regulatory considerations further limit real-world implementation. Consequently, while the findings provide valuable insights into risk–return trade-offs and diversification benefits, they should be interpreted as indicative rather than prescriptive, serving as a foundation for informed and disciplined portfolio decision-making rather than a rigid investment rule.